Assignment

Engineering Thermodynamics

1. (a) 1 kg of water at 293 K is brought into contact with a heat reservoir at 393 K. When the water has reached 393 K, find the entropy change of water, of the heat reservoir, and of the universe. (b) If the water is heated from 293 K to 393 K by first bringing it in contact with a reservoir at 343 K and then with a reservoir at 393 K, what will the entropy change of the universe be?
2. Calculate the decrease in exergy when 18 kg of water at 92°C mixes with 32 kg of water at 28°C in an insulated rigid tank at constant pressure. The surroundings are at a constant temperature of 12°C. Take cp = 4.2 kJ/kg·K for water (treat as incompressible liquid, constant specific volume).
   Determine:
   (a) The final equilibrium temperature of the mixture.
   (b) The entropy generation during the irreversible mixing process.
   (c) The decrease in exergy due to the mixing.
3. A fluid at 250 kPa and 350°C has a volume of 0.6 m^3. In a frictionless process at constant volume, the pressure changes to 120 kPa. Find the final temperature and the heat transferred if the fluid is steam.
4. A fuel having the formula $C_{10}H_{22}$ (Decane) is burnt with the chemically correct amount of air at atmospheric pressure. Write the balanced combustion equation assuming complete combustion to CO_2 and H_2O(g), and find the stoichiometric air-fuel ratio by mass
5. A three-stage air compressor takes in air at 1 bar, 300 K and delivers it at 50 bar. The maximum temperature in any stage is not to exceed 400 K. If the law of compression and expansion is $pV^{1.3}$ = constant, find the number of stages required for minimum power input if the delivery pressure is fixed at 50 bar. Also, estimate the power required per kg/s of air delivered and the heat rejected in each intercooler.